Neurological Physiotherapy — OSCE Prep

Cases 1 & 2 | Target: 70%+ (Very Good) | Clinical reasoning throughout

Case 1 — Mr. Okoro (Acute MCA Stroke) | Case 2 — Mr. Ibrahim (Cerebellar Ataxia)

Case 1 — Mr. Samuel Okoro

68 M | 5 days post-ischaemic stroke — Right MCA | Acute Stroke Unit, medically stable

Left hemiparesis (UL + face) · **Left-sided neglect** · **Pusher syndrome** · **Poor trunk control** · **2-person assist STS** · **Day 5 acute**

1 SUBJECTIVE ASSESSMENT

OPENING / CONSENT STATEMENT

> *"Good morning Mr. Okoro. My name is [your name] and I'm the physiotherapist looking after you today. I'm here to assess your movement and help you begin your recovery. Everything we do is completely voluntary — you can stop or rest at any time. I'll explain everything as we go. Is that okay with you?"*

⚠ **Positioning note:** Because of left-sided neglect I will initially position myself on Mr. Okoro's **right side** to ensure reliable communication, then deliberately engage the left side during assessment to characterise the neglect.

KEY SUBJECTIVE QUESTIONS + CLINICAL REASONING

Question I ask	Why — Clinical Reasoning
"Can you describe what you're feeling in your left arm and leg right now?"	Establishes the patient's own perception of motor loss and any sensory change (numbness, tingling). Also gauges his level of insight — anosognosia is common with right-hemisphere lesions and directly affects cooperation with rehab.
"Have you noticed any difficulty keeping yourself upright when you try to sit or move? Does it feel like you want to lean to one side?"	Screens for pusher syndrome and trunk control deficits from the patient's perspective. Understanding whether he perceives himself as upright when he is tilted helps me gauge the severity of his distorted subjective postural vertical — the core driver of pusher syndrome.
"Do you notice what's happening on your left side? Are you aware of your left arm, for example?"	Directly screens for left hemispatial neglect and the degree of insight. Neglect severity is one of the strongest predictors of poor functional outcome after stroke and fundamentally changes my treatment setup and cuing strategy.
"Do you have any pain — particularly in your left shoulder or arm?"	Hemiplegic shoulder subluxation and pain develop rapidly in a flaccid limb. Early shoulder pain predicts significantly worse upper limb recovery. I need to know so I can begin protective strategies immediately and alert the team.
"Have you had any difficulty swallowing food or drinks, or any changes to your speech?"	Right MCA lesions can affect cortical swallowing centres. Dysphagia carries an immediate aspiration risk — I must flag this to the speech and language therapist today. Dysphasia also changes how I communicate during therapy.
"Before all of this happened, how were you getting around — were you independent at home?"	Pre-morbid function is my recovery target. Knowing whether he was independently mobile, used any aids, or had other conditions gives me realistic goal anchors and helps determine discharge destination early.

RED FLAGS STOP & ACT

- **New or worsening neurological deficit** (e.g., rapidly increasing hemiplegia, new facial droop, sudden confusion): May indicate stroke extension or haemorrhagic transformation — cease physiotherapy immediately and escalate to the medical team.
- **Dysphagia / wet voice / coughing on fluids:** Immediate aspiration risk — refer urgently to SLT; do not allow oral intake without SLT clearance.
- **Cardiovascular instability** (BP >180/110 mmHg or <90/60 mmHg, resting HR >100, SpO_2 <94%): Physiotherapy is contraindicated — notify the nurse in charge and await medical review before any active rehabilitation.
- **Unilateral calf swelling, warmth, or redness in the hemiplegic leg:** Suspected DVT — mobilisation risks pulmonary embolism; urgent Doppler ultrasound and medical review required.
- **Severely altered consciousness or acute agitation:** Indicates possible raised intracranial pressure or metabolic complication — unsafe for active therapy; inform the nurse in charge immediately.

PROBLEM LIST

Physical Impairments

- Left UL and LL hemiparesis (flaccid acutely)
- Left lower facial weakness (UMN pattern)
- Trunk control deficit — unable to sit unsupported
- Pusher syndrome — active lateral pushing toward left
- Left hemispatial neglect
- Presumed left sensory impairment (proprioception, light touch)
- Risk of hemiplegic shoulder subluxation

Functional Activity Limitations

- Requires 2-person assist for sit-to-stand
- Unable to bear weight through left LL safely
- Fully dependent for all transfers
- Dependent for all personal ADLs
- Unsafe to mobilise unassisted

PSYCHOSOCIAL / YELLOW FLAGS

- **Acute emotional distress / fear:** A sudden devastating event 5 days ago — expect high levels of fear and anxiety about the future. I will screen for low mood using a validated tool (e.g., PHQ-2) and refer to psychology if indicated.
- **Reduced insight / anosognosia:** Right-hemisphere neglect frequently co-exists with poor awareness of deficits, which reduces participation and increases risk-taking. Careful, non-confrontational education is essential.
- **Social support and discharge planning:** Begin exploring home situation, family support, and social circumstances from Day 1 — acute stroke unit length of stay is short and discharge planning needs to start now.
- **Cultural and language considerations:** Confirm preferred language; use a trained medical interpreter if needed rather than family members, particularly for consent and goal-setting conversations.

2 PATHOPHYSIOLOGY OF WORKING DIAGNOSIS

WHAT IS HAPPENING NEUROLOGICALLY

Mr. Okoro has sustained an ischaemic infarct in the territory of the **right middle cerebral artery (MCA)**, which supplies the lateral convexity of the cerebral hemisphere including the primary motor cortex (precentral gyrus), primary somatosensory cortex, and parietal association areas on the right side. Occlusion of the right MCA results in infarction of the **corticospinal tract fibres** subserving the contralateral (left) body, producing upper motor neuron hemiparesis. In the acute phase — the first days to weeks — the phenomenon of **diaschisis** causes widespread suppression of neuronal activity in areas connected to but remote from the infarcted zone, resulting in the **flaccid hypotonia** seen initially rather than the spasticity that emerges later as diaschisis resolves. Damage to the **right parietal cortex and its fronto-parietal attention network** disrupts the right hemisphere's dominant role in bilateral spatial attention, producing left hemispatial neglect — a failure to attend to, represent, or respond to stimuli on the contralesional side of space. **Pusher syndrome** arises from infarction of the posterior thalamus or insula — regions responsible for processing the subjective postural vertical (SPV): the patient's brain incorrectly signals that they are upright when they are tilted toward the hemiplegic side, and they actively push against support to restore their distorted SPV. Neuroplasticity — encompassing synaptic unmasking, axonal sprouting, and cortical map reorganisation — begins within hours of the infarct and is the biological substrate that physiotherapy exploits through high-repetition, task-specific practice to drive functional recovery.

TYPICAL SIGNS & SYMPTOMS EXPLAINED BY PATHOPHYSIOLOGY

Sign / Symptom	Pathophysiological Explanation
Left-sided hemiparesis (UL > LL)	Right motor cortex / corticospinal tract infarction. UL fibres occupy a more lateral and superior position in MCA territory, hence disproportionately affected by MCA occlusion compared with the more medially represented LL fibres (ACA territory).
Left lower facial weakness	UMN lesion — contralateral lower face only, because the forehead (frontalis) has bilateral cortical representation. Classic distinction from LMN facial palsy which affects all ipsilateral facial muscles.
Left hemispatial neglect	Right hemisphere parietal / fronto-parietal network damage. The right hemisphere normally attends to both sides of space; the left attends only to the right. With right hemisphere damage, nothing compensates for left-sided inattention.
Pusher syndrome (pushing toward left)	Thalamo-cortical processing of SPV is disrupted. The patient's internal body-vertical is shifted leftward; they perceive themselves as falling to the right and push left (toward hemiplegia) to "correct" this.
Flaccid tone acutely / poor trunk control	Acute diaschisis suppresses cortical drive to the affected side including trunk musculature. Corticospinal facilitation of postural tone is lost, leaving the trunk without active anti-gravity control.

WHY THIS PATIENT FITS THE DIAGNOSIS

Mr. Okoro's presentation is a textbook right MCA syndrome. **Contralateral (left-sided) hemiparesis with UL predominance** localises to the right motor cortex, in the MCA territory. **Left hemispatial neglect** is strongly lateralised to the right hemisphere — it is far more severe and persistent after right-sided lesions — confirming right parietal involvement. **Pusher syndrome** in conjunction with neglect is characteristic of right-hemisphere strokes involving the thalamic and insular SPV processing networks. The **flaccid tone at 5 days** is consistent with the acute diaschisis phase of recovery. The requirement for **2-person assist** reflects the compounded challenge of hemiparesis, trunk instability, and the unpredictable nature of pusher behaviour during loaded tasks.

3 DIFFERENTIAL DIAGNOSES

Primary Diagnosis: Right MCA Ischaemic Stroke with Left Hemiparesis, Hemispatial Neglect & Pusher Syndrome
Supported by: acute onset, neuroimaging confirmation, classic MCA-territory deficits — contralateral motor/sensory loss, neglect, and pusher syndrome.

DIFFERENTIAL 1 — INTRACEREBRAL HAEMORRHAGE (ICH)

Supporting features:

- Sudden onset severe neurological deficit — identical presentation clinically
- Affects similar cortical and subcortical regions
- Produces hemiparesis, hemisensory loss, and consciousness disturbance

Why I favour ischaemic stroke over ICH: *The diagnosis of ischaemic stroke is confirmed by neuroimaging (CT/MRI). Furthermore, ischaemic strokes are approximately five times more common than haemorrhagic strokes. Mr. Okoro is described as "medically stable" at Day 5, which is more consistent with a surviving ischaemic event; large ICH carries higher acute mortality and instability. Anticoagulant management also differs — important context for physiotherapy safety planning.*

DIFFERENTIAL 2 — SUBDURAL HAEMATOMA (SDH)

Supporting features:

- Can produce contralateral hemiparesis in a 68-year-old via cortical compression
- More common in older adults (cerebral atrophy stretches bridging veins)
- Fluctuating consciousness and headache may accompany motor signs

Why I favour MCA stroke over SDH: *SDH typically presents with a subacute or fluctuating course and is usually associated with head trauma or anticoagulant therapy. The highly specific constellation of left neglect, left hemiparesis, and pusher syndrome in an acute vascular pattern strongly favours a focal cortical infarct. Neuroimaging has confirmed this, excluding SDH. SDH would also produce more diffuse cortical compression effects rather than the focused MCA-territory deficit profile seen here.*

4 OBJECTIVE ASSESSMENT — SPECIAL TESTS

KEY NEUROLOGICAL OBSERVATIONS BEFORE TESTING

- **Tone:** Passive movement of UL and LL — expect flaccid / hypotonic acutely. Use Modified Ashworth Scale. Note any early spasticity emerging in wrist flexors or ankle plantar flexors.
- **Posture:** In supported sitting — observe direction and magnitude of trunk lean, active pushing against the plinth toward the left, and head position. Does he perceive himself as upright?
- **Neglect:** Observe spontaneous attention — does he orient only to the right? Does he fail to notice or reach for items on the left? Does he leave the left arm in unsafe positions?
- **Coordination:** Difficult to assess separately from weakness acutely — note quality and trajectory of any voluntary movement attempted in the UL.
- **Gait / weight-bearing:** Not applicable yet — note the quality of any weight-shifting attempted during STS practice. Does he load the left leg when cued?

Test 1 — Scale for Contraversive Pushing (SCP)

What it measures: The severity of pusher syndrome — specifically, the degree to which the patient spontaneously pushes toward the hemiparetic side and actively resists passive correction back to midline.

Why I choose it: *Pusher syndrome is the primary safety barrier for this patient — it is the reason 2-person assist is required and the feature that will most delay his progression to standing and walking. The SCP is the gold-standard validated tool for quantifying this. I need a baseline score to communicate severity to the MDT, justify physiotherapy goals, and measure change over time. Without this measurement I cannot demonstrate whether my treatment is working.*

How to perform:

1. Observe the patient in sitting (and standing if tolerated) without manual correction from the therapist.
2. **Posture subscale (0–1):** Is there spontaneous lateral lean toward the affected side? Score 0 = no lean; 0.25 = slight lean; 0.75 = moderate; 1 = extreme lean.
3. **Extension subscale (0–2):** Does the patient use the non-paretic arm or leg to actively push (extend and abduct)? Score 0 = no; 1 = in one position; 2 = in both sitting and standing.
4. **Resistance subscale (0–1):** When the therapist attempts to passively correct the patient toward true midline, does the patient resist? Score 0 = no resistance; 1 = resistance present.
5. Sum all subscale scores. Total range 0–4; score ≥1 = pusher syndrome present.

Positive finding: SCP ≥1. Clinical meaning: the patient is actively pushing toward the hemiplegic side and resisting passive correction — their subjective postural vertical is severely distorted. Higher scores (3–4) predict significantly longer inpatient stays. This score drives the priority of my entire initial treatment plan.

Test 2 — Line Bisection Test (Hemispatial Neglect Screen)

What it measures: The presence and severity of left hemispatial neglect — specifically, the patient's ability to allocate attention equally to both halves of extrapersonal space.

Why I choose it: *Neglect independently predicts functional outcome and fall risk post-stroke. It directly determines how I set up the therapy environment, where I position myself and meaningful objects, and what compensatory strategies I teach the patient and carers. Documenting severity at baseline gives the MDT a shared reference and allows me to track spontaneous recovery of attention over the admission.*

How to perform:

1. Place an A4 sheet with a horizontal line (200 mm long) centred on the page in front of the patient at their midline.
2. Instruct: "Please mark the middle of this line with your pen."
3. Measure deviation of their mark from the true centre in millimetres; rightward deviation indicates left neglect.
4. Repeat with 3 lines of varying lengths and average the deviation.
5. Supplementary: administer the **Star Cancellation Test** — patient circles all small stars on a busy page; left-sided omissions confirm neglect.

Positive finding: Mark placed significantly to the right of centre (>6 mm rightward deviation). Left-sided omissions on Star Cancellation. Clinical meaning: confirms left hemispatial neglect and quantifies severity, guiding cuing strategy selection and environmental setup throughout all therapy sessions.

Test 3 — Trunk Control Test (TCT)

What it measures: Sitting and lying-to-sitting postural control across four tasks — rolling to the weak side, rolling to the strong side, sitting up from lying, and static sitting balance. Scored 0–100; <50 at one week post-stroke strongly predicts dependency at discharge.

Why I choose it: *Trunk control is the prerequisite for every functional recovery goal — sitting, standing, walking, and ADLs all build on trunk stability. At Day 5, the TCT gives me a baseline prediction of functional trajectory. A score <50 tells me and the team that this patient is likely to need inpatient rehabilitation — critical information for bed planning. It also identifies the specific trunk tasks to target in my early sessions.*

How to perform:

1. Position patient supine on the bed (ensure safety side rails lowered with assistance standing by).
2. **Roll to weak side:** Score 0 (unable), 12 (assisted), 25 (independent).
3. **Roll to strong side:** Same scoring.
4. **Sit up from lying:** Same scoring.
5. **Sitting balance:** Can patient sit without support for ≥30 seconds without falling? Score 0 (unable) or 25 (achieved).
6. Total = 0–100.

Positive / abnormal finding: Expected to be very low (<50) given the 2-person assist requirement. Score <50 at Day 5–7 predicts functional dependency at discharge and confirms that trunk control rehabilitation must be my number one priority in weeks 1–2.

OUTCOME MEASURES

Measure	Purpose	When
Barthel Index / FIM	Functional independence across ADL domains; standard acute stroke measure	Admission, weekly, discharge
Trunk Control Test (TCT)	Sitting balance — early outcome predictor	Baseline and 2-weekly
Scale for Contraversive Pushing (SCP)	Pusher syndrome severity and progress	Baseline and each session
NIHSS (medical)	Stroke severity context — know the patient's score	On admission (medical team)

5 TREATMENT PLAN

INITIAL PHASE — WEEKS 1–4 (ACUTE INPATIENT)

Core Neurophysio Interventions:

- **Early mobilisation (upright positioning and sitting out within 48–72 hours of stroke, medically stable):** *Evidence supports early upright positioning to prevent deconditioning, DVT, and pneumonia. However, the AVERT trial cautions against very frequent high-dose very early mobility in the first 24 hours — I will follow the unit's protocol and escalate cautiously. From Day 2–3, sitting out of bed in a supported chair is the minimum daily goal.*
- **Pusher syndrome treatment via vertical visual reference:** During all sitting and standing practice I will use a vertical reference (door frame, vertical line on a mirror, or a plumb line) and instruct the patient to align his trunk with it. *Research by Karnath et al. demonstrates that visual vertical references are the most effective intervention for pusher syndrome — they bypass the distorted internal SPV by providing an accurate external vertical cue. Crucially, I will not physically push him back to midline against his resistance, as this increases pushing behaviour.*
- **Left neglect management — bottom-up cuing and environmental modification:** Red anchor strip on the left margin of all written materials; approach from right initially; place meaningful objects on the left during therapy; tactile cuing to left UL; consider prism adaptation therapy (liaise with OT). *Bottom-up attentional cuing temporarily overcomes neglect-related inattention and, with repetition, begins to drive neuroplastic reorganisation of the right hemisphere's attentional network.*
- **Hemiplegic shoulder protection from Day 1:** Correct positioning in lying (arm on pillow), sitting (arm on arm rest or lapboard), and during all transfers (no traction on the UL). Apply a sling for vertical positions if subluxation confirmed. *Shoulder subluxation and subsequent pain are almost universal in flaccid hemiplegia if left unmanaged. Shoulder pain is one of the strongest negative predictors of UL recovery — prevention is far more effective than treatment once pain is established.*
- **Daily passive and active-assisted range of motion (PROM / AAROM):** Full PROM to left UL and LL, particularly shoulder, elbow, wrist, hip, knee, and ankle. *Prevents soft tissue contracture in a flaccid limb. Contracture is a secondary complication that becomes a permanent barrier to recovery — completely avoidable with early, consistent ROM. Also provides sensory input that may support cortical reorganisation.*

Exercise 1 — Supported Sitting: Trunk Midline Orientation Training

How to perform: Patient sits at the edge of the plinth or in a supportive chair with feet flat on the floor. A vertical reference (mirror with vertical line, door frame, or plumb line) is placed directly in front. Therapist provides verbal cuing: "Look at the vertical line — align your shoulders with it." Light manual facilitation at the pelvis guides the trunk toward true midline. Hold for 10–30 seconds. Rest and repeat 5–8 times per session.

Purpose: Directly targets pusher syndrome by recalibrating the SPV using external visual input. Also trains trunk anti-gravity control as the foundational skill for all subsequent functional tasks.

🔽 **REGRESSION**
30–45° reclined in bed (reduces trunk demand). Focus only on head and cervical alignment toward vertical reference. Full manual support at trunk. Very short holds (5 seconds). Eyes open always.

🔼 **PROGRESSION**
Remove external vertical reference (patient uses internal cue only). Add brief eyes-closed intervals. Reduce manual support to fingertip contact only. Add reaching tasks toward left side — combines neglect and trunk training. Extend hold duration toward 60+ seconds.

Exercise 2 — Sit-to-Stand (STS) Practice with Facilitation

How to perform: Begin with plinth at maximum height to reduce the STS demand. Feet flat on floor hip-width apart, left foot positioned carefully (neglect risk). Therapist uses key-point control: one hand at the patient's affected knee (to prevent knee collapse and guide weightbearing), one hand at pelvis (to facilitate forward lean). Verbal cue: "Lean your nose toward your toes, push through both feet equally." Use the vertical reference to cue upright alignment on the way up. Aim for 3–5 repetitions per session, 2 sessions per day.

Purpose: Fundamental functional task — practising it repetitively drives neuroplasticity along the corticospinal tract to the affected LL. Forces symmetric weightbearing, targets pusher syndrome in a real functional context, and provides the greatest carryover to ADL independence.

🔽 **REGRESSION**
Use a tilt table for partial standing weight-bearing with zero trunk demand. Or practice weight shift only — leaning forward from sitting without completing the stand. High plinth, 2-person assist, full support. Focus on loading the left foot rather than completing the STS task.

🔼 **PROGRESSION**
Lower the plinth height progressively. Reduce from 2-person to 1-person assist. Introduce brief standing hold (5–10 seconds). Add stepping in place once standing is achieved with control. Then begin gait re-education with appropriate assistive device.

SAFETY CONSIDERATIONS

- Maintain 2-person assist protocol until SCP score is consistently <1 and midline control in standing is reliable.
- Monitor BP, HR, and SpO_2 before and after each session — haemodynamic instability is common in the first week post-stroke.
- Protect the hemiplegic shoulder in every session and transfer — no traction, no UL pulling; educate all staff and carers in safe handling.
- Ensure the left arm is always visible and safely supported — neglect means the patient will not feel or notice if his arm is trapped, compressed, or hanging unsupported.
- Keep sessions brief (15–20 minutes) and frequent — the acute post-stroke brain fatigues rapidly; short frequent sessions outperform single long ones in this phase.
- Bed at lowest position, call bell within reach on the right side, bed rails up when not in therapy, carer aware of pushing behaviour and fall risk.

PATIENT & CARER EDUCATION — KEY MESSAGES

1. **"The brain can rewire — but it needs practice."** Explain neuroplasticity in plain terms: the more Mr. Okoro attempts movement, the more the brain reorganises. Every therapy session and every attempt between sessions counts. Recovery is not passive.
2. **Left neglect safety:** Educate carers to always check that the left arm is supported and visible, to approach from the right initially when speaking to Mr. Okoro, and to regularly prompt him to look to his left — e.g., "Check your left side."
3. **Shoulder protection technique:** Demonstrate safe handling to carers — support under the elbow, never lift or pull by the arm, correct positioning in bed with arm on pillow. Show what NOT to do (dragging by the wrist during transfers).
4. **Understanding pusher syndrome:** Reassure the patient and family that the pushing is not deliberate or a sign of poor effort — his brain is giving him incorrect information about his position in space. The treatment aim is to help his brain recalibrate using what he can see, not to fight against the push.

PROGRESSION CRITERIA — MEASURABLE TARGETS

- SCP score consistently <1 → progress to 1-person assist and begin standing tolerance training
- TCT score >50/100 → progress to standing and stepping practice
- Sustained unsupported sitting ≥30 seconds → begin standing transfers independently
- Patient achieves 1-person assist or supervised STS with consistent left foot loading → step down to rehabilitation ward
- Barthel Index / FIM motor subscale improves ≥10 points from admission → demonstrates meaningful functional progress

LONG-TERM PLAN — WEEKS 4–12+ (REHABILITATION WARD / COMMUNITY)

- **Transfer to inpatient stroke rehabilitation ward** for intensive MDT rehabilitation once meeting progression criteria above.
- **Gait re-education:** Parallel bars → wheeled walking frame → single-point stick, with ongoing focus on symmetric gait pattern, left limb control, and ankle stability (consider AFO if foot drop emerges with spasticity).
- **Upper limb rehabilitation:** Task-specific training when voluntary movement returns; mirror therapy continuation; CIMT (constraint-induced movement therapy) in later stages when at least 10° of wrist extension present.
- **Spasticity management:** Monitor for spasticity emerging in wrist and finger flexors, ankle plantar flexors. Introduce positioning, stretching, and splinting; liaise with medical team regarding Botulinum toxin if clinically indicated.
- **Community discharge:** Pre-discharge home visit; environmental assessment (grab rails, shower seat, threshold ramps); community physiotherapy and OT referral; carer training programme.
- **Discharge criteria:** Safe independent or supervised mobility for home environment, carer competent in all safe handling, community supports confirmed, secondary prevention plan in place.

EXAMINER DEMONSTRATION — WHAT I WOULD SHOW IN THE ROOM

1. **STS facilitation technique:** Demonstrate key-point control at the knee and pelvis while verbally cuing symmetric weightbearing and forward lean. Show how I position the vertical reference and use it to address pusher syndrome simultaneously during the movement.
2. **Safe hemiplegic shoulder handling during a transfer:** Demonstrate correct support under the elbow, correct arm positioning in sitting, and clearly narrate what must be avoided (traction on the arm). Show the carer exactly how to assist safely.